

Reproduction quickstart

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Everything needed to re-run the results in the mentor-review draft, in the order a reviewer would want them.

1. Get the source

```
git clone https://github.com/uskutsav-cpu/selfdual-5form-invariants
cd selfdual-5form-invariants
git switch publication/jhep-mentor-draft
```

2. Environment

```
python3 -m venv .venv
.venv/bin/python3 -m pip install -r requirements.txt
```

Dependencies are NumPy, pynauty and pytest.

Two notes that will otherwise cost you time:

- `opt_einsum` is imported by the spinor code but is **optional**, is **not** in `requirements.txt`, and was **not installed** in the environment that produced the certified results. The built-in contraction ordering was used. Installing it should change speed and not values, but that was not certified both ways.
- If you use the repository's committed `.venv`, its `pip` shebang points at a path that no longer exists. Use `.venv/bin/python3 -m pip` rather than `.venv/bin/pip`.

3. Test suites

```
.venv/bin/python3 -m pytest tests -q
.venv/bin/python3 -m pytest spinor_trace_bridge/tests -q
```

Expected: **254** tensor tests and **86** bridge tests, no failures.

4. The exact rational certificates

```
.venv/bin/python3 scripts/d10_characteristic_zero.py
.venv/bin/python3 scripts/b10_p10_characteristic_zero.py
```

Expected:

quantity	value
dim_Q A10	14
dim_Q D10	11
dim_Q Q10	3
dim_Q(B10 P10)	1

The first script lifts 9 non-integral rows by CRT across 5 primes, validates at held-out prime 32771, and runs the closure to a fixed point in `Fraction` arithmetic (3 sweeps). The second lifts all 12 published structures across 7 primes, validates at held-out prime 32783, and re-verifies the generator identity at fresh prime 32869 on 6 fresh samples.

5. The rank matrix

```
.venv/bin/python3 scripts/aggregate_rank_matrix.py --self-test
```

Read-only with respect to the cells and order-independent, so it is safe to run at any time. Expected: 15 cells, rank 81 in every one, 81 stable pivot rows and 81 stable pivot columns, 0 unstable.

Caution. Individual cells may be run in parallel but must not share a row cache. Two processes writing one cache is a failure this project has actually experienced.

6. Rebuild the manuscript

```
.venv/bin/python3 manuscript/scripts/make_numbers.py
.venv/bin/python3 manuscript/jhep/scripts/make_figures.py
.venv/bin/python3 manuscript/jhep/scripts/make_tables.py
.venv/bin/python3 manuscript/jhep/scripts/check_draft.py
cd manuscript/jhep && tectonic -X compile main.tex
```

The macro, figure and table steps must precede the compile: the manuscript contains no scientific value of its own, only macros that expand to values read from artifacts.

`check_draft.py` runs 52 gates covering LaTeX health, citation coverage, placeholder discipline, and the wording restrictions (no unscoped completeness or canonicity claims, rank 81 scoped to the selected family, degree-eight claim qualified, degree-twelve scope stated).

Expected: **52 gates passed, 0 failed**, and a 41-page PDF with zero overfull boxes, zero underfull boxes, zero undefined citations and zero undefined references.

7. A note on TeX

The draft compiles with **Tectonic**. Because `jheppub.sty` requests `hyperref` with `pdfa=true` — which selects the pdfTeX driver and fails under Tectonic’s XeTeX engine — `main.tex` passes `xetex` to `hyperref` before the class is loaded. The official JHEP style file is not modified. Compiling with pdfLaTeX instead should also work and does not need that line.

8. Determinism

Figures are byte-identical across runs; matplotlib’s PDF creation timestamp is suppressed explicitly. Archives use a fixed epoch and normalised member metadata. If you regenerate the figures and the hashes differ, that is a finding, not noise.